Q1:
So, I asked my professor on what does a final year project should look like? should it be from a IEEE paper?
He answered that it maybe newspaper could be a good source of ideas. When I read this news, I thought that I can build a system to solve this problem as I recently completed by OOAD NPTEL course in which I designed multiple systems.

So, I found it as an interesting thought exercise. We had about 1.5 month before the next academic year starts so I thought of giving it a try to see whether it makes sense to solve this problem.

Another reason to solve this was that I was deeply frustrated for atleast 3 years (2015-18) with the poor public city bus transport of Nashik. It was my daily routine to wait for .5 - 1.5 hr for the bus to arrive. We didn't had a digital system to know the ETA of the bus. Also, the bus frequency was less. So, one bus going from our college couldn't take all the waiting students. So, it was a irritating experience everyday to keep on waiting for your bus to arrive.

Q2:
OOAD gave me a framwork/perspective to see things from a systems lens. It helped me structure my thoughts and gave me the confidence that I could design any system.
Without OOAD Nashik Bus Assistant might have happened but we couldn't have made such progress within the deadline. The development would have been very unstructured and messy.

Q3:
I assumed the problem was that the city bus governing bodies didn't hear the needs of its daily commuters. Because, the bus timetables were erratic and never on schedule. We lacked a digital system that could provide real-time bus information.

Q4:
We would talk to daily commuters all day and see whether the idea of product made sense. We checked with the depot managers on how do they run the bus fleets. We collected data from MSRTC offices, did studies on successful city bus systems of different cities like, Indore, Mysore, Bengaluru.

Q5:
I realized that this was the same issue that was there in my home town as well. And there were multiple cities in Maharashtra that had the same issue. That means the buses used to operate in the same orthodox, non-digital way. And there are still commuter waiting times that are high because there is no digital system in place. So, we thought that if we could build a system for Nashik, and if we could prove our product relevance, then we might be able to scale it to different cities within the Maharashtra, and then scale as we go. So we saw this as a very good startup opportunity.

Q6:

Because I faced this issue every day. Like I used to wait for two hours, three hours just for the buses to arrive so that I could reach my college and then way back at my home. So I was particularly very disappointed with the poor public transportation service in the city. So I thought that, I have the relevant skill set to solve this problem, so let's give it a shot.

Q1:
So we started to design the system based on our understanding of the commuter pain points, the depot manager's inputs, and the relevant stakeholder conversations that we had. But we thought that this could be a live tracking GPS-enabled real-time system that could help the commuter with the real time updated information. But this idea became flawed because we did not have either the GPS or the bus stop coordinates for the entire city. So we could not proceed ahead with the kind of design that we initially had in mind because we did not have all these things. So this became our major pivot.

Q2:
So when we were thinking of a GPS-enabled system, people were interested in making the map of the city and the highlighted bus routes based on the user search. People were like that we will work on getting the bus stop coordinates and then plotting it on the map so that the user can see the direction and the route of the bus, while the others were interested in the backend cloud technology that we should be focusing on to make the system work offline. These were different aspects. But we did not have either the coordinates or the understanding of the cloud technologies that we could be using for our MVP. So instead of directly implementing it, we thought that first let us inquire with the MSRTC officials to check whether we have the bus stop coordinates. And I also asked some of the members to research on the kind of cloud technologies that we could use as a backend for this app. Once we have the answers to these questions, only then we decided on our next steps so that everyone is online.

Q3:
The assumption that we will get all the data that we want to build this product turned out to be wrong because we just assumed that all this data is available to us.

Q4:
My first impression was that it was a very traditional legacy organisation where people used to work on pen and paper and very few used computers. Also, it was mostly based on intuition rather than some solid research. The people there were very process-oriented and there was little room for creativity and innovative solutions and technology that could help to improve the efficiency of the system.

Q5:
Before building NBA (Nashik Bus Assistant), if I were to go in that phase again, I would talk to the MSRTC officials and the Nasik Municipal Corporation to check whether there is a feasibility for us to come up with a system that could be accepted by both the parties and whether building

such a system would get the support from the relevant organizations and ground-staff. That means, I'll check whether the incentives of customer, business and technology folks align.

Purpose

Build products that improve people's lives by combining technology, user empathy, and business thinking.

Core Beliefs

- Great products start with understanding people.
- Constraints inspire better product decisions.
- Product Managers create alignment across technology, business, and users.
- Shipping is important, but solving the right problem is more important.

Strengths

- Systems thinking.
- Structured problem solving.
- Initiative.
- Leadership.
- Customer empathy.

I enjoy solving large-scale, systemic problems where technology can meaningfully improve everyday experiences for many people. I'm motivated by problems where improving a system creates meaningful benefits for large groups of people. I enjoy understanding broken systems and designing products that make them work better for the people who depend on them. I believe Product Management is fundamentally about reducing uncertainty. Someone who enjoys taking ambiguous, real-world problems and systematically reducing uncertainty until a practical product can be built.

Portfolio Strategy (Version 0.1)

1. Purpose

Objective

Build a recruiter-ready Product Management portfolio that demonstrates my product thinking, leadership, customer empathy, structured decision-making, and ability to build products from opportunity discovery to execution.

Primary Goal

Position myself as an aspiring Product Manager who can solve meaningful, large-scale problems by combining technology, business thinking, and customer understanding.

---

2. Target Audience

Primary

- Product Recruiters
- Hiring Managers
- Senior Product Managers
- Product Directors

Secondary

- Fellow Product Managers
- Startup Founders
- Engineering Managers

Target Companies (initial focus)

- Google
- Microsoft
- Amazon
- Atlassian
- Razorpay
- CRED
- Swiggy

- Uber
- Airbnb
- Other technology companies hiring PMs

(We can refine this later.)

---

3. Desired Recruiter Impression

When someone finishes reading my portfolio, I want them to think:

- This person deeply understands users.
- This person approaches problems with structured thinking and sound judgment.
- This person can align people and lead cross-functional teams toward meaningful outcomes.

---

4. Personal Mission

Use technology to solve meaningful, large-scale problems by understanding users, business needs, and the broader ecosystem in which products operate.

---

5. Why Product Management?

I believe technology has the power to positively influence people's lives at scale.

While studying Computer Science, I realized that technology alone is not enough to build successful products. Understanding customers, business objectives, and execution is equally important.

Product Management sits at the intersection of these disciplines, which is why I chose to pursue it.

---

6. Current Product Philosophy (Working Draft)

I believe:

- Great products begin with a deep understanding of people.
- Technology alone does not create successful products.
- Successful products emerge when user needs, technology, business incentives, and the surrounding ecosystem are aligned.
- Constraints often lead to better product decisions.
- Product Managers should understand the system before proposing solutions.

> **Note:** This section is intentionally a working draft. It will evolve as we complete the case studies.

---

7. Personal Strengths (Evidence Pending)

Current hypothesis:

- Systems Thinking
- Customer Empathy
- Structured Problem Solving
- Initiative
- Leadership
- Cross-functional Collaboration

We'll replace these with evidence-backed statements as we complete the Nashik Bus Assistant case study.

---

8. Long-term PM Interests

Current areas of interest:

- Consumer Products
- Growth
- Platform

This will evolve as I build more projects.

---

9. Differentiators

Working hypothesis:

- I naturally identify overlooked real-world problems.
- I enjoy understanding complex systems before proposing solutions.
- I take initiative to build products even without formal authority.
- I am motivated by solving problems that create meaningful impact for large groups of people.

Again, we'll validate these with evidence rather than leaving them as claims.

---

10. Portfolio Principles

This might become the most important section.

Every page we publish must satisfy these principles:

Authenticity

Never exaggerate outcomes or claim impact that wasn't measured.

Evidence First

Every important claim should be backed by:

- project artifacts,
- user research,
- documented decisions,
- or personal experience.

Product Thinking over Technology

Focus on:

- decisions,
- trade-offs,
- prioritization,
- user understanding,
- business context,

rather than implementation details.

Reflection

Every case study should answer:

> What would I do differently today?

Recruiter Readability

Every page should answer:

- Why did this product exist?
- What problem was solved?
- Why was this approach chosen?
- What trade-offs were made?
- What happened?
- What did I learn?

---

11. Current Portfolio Roadmap

Phase 1

Build one flagship case study.

✅ Nashik Bus Assistant

Phase 2

Publish the portfolio.

Phase 3

Add additional case studies.

Phase 4

Product writing.

Phase 5

Continuous improvement.

Why this problem?
Why buses?
Why technology?
GPS vs. heuristic ETA.
Data availability assumptions.
Organizational feasibility.
What I'd do differently today.

Decision 1

Should we solve this?

↓

Decision 2

What problem are we actually solving?

↓

Decision 3

Should it be GPS?

↓

Decision 4

How do we build without GPS?

↓

Decision 5

Can the organization adopt this?

↓

Reflection

Why I build products

> I believe technology can improve people's lives at scale, but only when it solves meaningful problems within the realities of business and society.

How I think

> I start by understanding the system before proposing solutions. Products are rarely isolated; they exist within a network of users, organizations, incentives, regulations, and market dynamics.

What I learned from Nashik Bus Assistant

> Building a technically sound product isn't enough. A product succeeds only when technology, user needs, organizational incentives, and the broader operating environment are aligned.

My role as a Product Manager

> My responsibility isn't just to define features. It's to understand the ecosystem, identify the right problems to solve, and align stakeholders around solutions that are both valuable and feasible.

Today I would first validate whether customer incentives, business incentives and technology constraints align before building the product.

Home

About

Product Philosophy

Case Studies

Resume

Contact

1. Hero / Product Snapshot

Nashik Bus Assistant

> **Designing a low-cost digital transit system for a city struggling with unreliable public transportation**

Immediately underneath:

Role	Product Lead / Product Manager
Team	7 members
Timeline	May 2018 – May 2019
Domain	Public Mobility / Civic Tech
Stakeholders	MSRTC, NMC, commuters, drivers, conductors, depot controllers
Scale of problem	70,000+ daily commuters
Status	Deployable MVP; public launch blocked by organizational alignment

This section is essential. A recruiter should understand the project in ~20 seconds.

---

2. The Story: Why I Built This

This should be your hook.

A newspaper headline turned a personal frustration into a product opportunity

Tell the story:

- You were taking OOAD.
- You asked your professor what makes a good project.
- He suggested looking at real-world problems/news.
- You read about Nashik's struggling bus service.
- You had personally experienced 0.5–1.5+ hour waits for years.

- You wondered whether technology could solve part of the problem.
- You had ~1.5 months before the next academic year and decided to investigate.

This is important because it establishes **intrinsic motivation + problem discovery**, rather than "college assigned me an app."

---

3. Business & Market Context

This is one of the most important sections and one of the things that differentiates NBA from a normal student project.

Explain the ecosystem you were entering:

MSRTC's problem

- City bus operations were loss-making.
- Fleet reductions were occurring.
- The organization wanted to reduce its financial burden.

NMC's position

- NMC was being pushed toward taking responsibility for city transportation.
- There was an ongoing institutional/ownership transition.

Commuter's problem

- Low bus frequency.
- Long waiting times.
- Unpredictable schedules.
- No digital visibility into bus arrival information.

The strategic tension

Show this visually:

Commuter needs
↑
Better availability + predictability

MSRTC incentives
↓
Reduce losses / eventually exit city operations

NMC incentives
↓
Take responsibility but manage financial/operational risk

Technology opportunity
→ Improve information + scheduling at relatively low cost

This is critical because **this tension ultimately explains the adoption problem.**

---

4. Problem Discovery

Don't immediately say:

> “We decided to build a bus-tracking app.”

Show how your understanding evolved.

Initial problem hypothesis

You initially believed:

> Commuters lacked reliable information because bus schedules were erratic and there was no digital system for real-time information.

Then explain how you investigated it.

Discovery activities

You:

- spoke with daily commuters,
- interacted with depot managers,
- spoke with drivers/conductors/controllers,
- collected operational data from MSRTC,
- studied other Indian city transport systems,
- examined how fleets and timetables were managed.

This is where we demonstrate **actual product discovery**, not framework knowledge.

---

5. What We Learned

This should be a distinct section.

Don't just say “research showed X.”

Show the **insights**.

For example:

Insight 1 — The commuter problem wasn't simply lack of buses

More predictable information could reduce uncertainty even when fleet capacity couldn't immediately change.

Insight 2 — Operational data was fragmented

The product depended heavily on timetable and route information controlled by the transport organization.

Insight 3 — Infrastructure constrained the ideal solution

GPS and comprehensive bus-stop coordinates weren't available.

Insight 4 — The product had multiple users

You weren't designing only for commuters.

There were also operational stakeholders.

Insight 5 — Adoption depended on institutional incentives

This became your biggest learning.

---

6. Users & Stakeholders

This deserves its own visual.

Primary user

Daily commuter

Needs:

- know which bus to take,
- know when it should arrive,
- reduce uncertainty,
- plan journey.

Operational users

Depot controller

Needs:

- manage schedules,
- understand route operations,
- update information.

Driver / conductor

Needs:

- communicate operational timing information.

Institutional stakeholders

- MSRTC
- NMC
- Smart City stakeholders

Then show the relationship between them.

This is a strong opportunity to demonstrate that you understood a **multi-sided product ecosystem**.

---

7. Product Vision

One concise statement.

Something like:

> **Make city bus transportation more predictable for commuters while giving transport operators a low-cost digital system to manage and communicate service information.**

Then explain your longer-term ambition:

> Prove the model in Nashik and potentially scale it to other cities facing similar public transport challenges.

Your own recollection that you saw this as potentially scalable across Maharashtra is valuable here.

---

8. Solution Exploration

This is where we show that **you didn't start with the final solution.**

I'd create a visual:

Initial concept
GPS-enabled real-time tracking
↓
Constraint discovery
No GPS infrastructure
No complete stop coordinates
Budget constraints
↓
Pivot
Heuristic ETA + timetable-based system
↓
MVP
Commuter + operations platform

This is potentially **the strongest visual in the case study.**

---

9. The Key Product Pivot

Give this its own section.

We couldn't afford the ideal solution. So we redesigned the problem.

Explain:

Original assumption

GPS-enabled buses → real-time location → accurate ETA.

Reality

- GPS unavailable.
- Stop coordinates unavailable.
- Budget constraints.
- Limited technical/infrastructure readiness.

Decision

Build an ETA system using available operational/timetable data and heuristic logic.

Trade-off

GPS approach	Heuristic approach
Higher accuracy potential	Lower accuracy
Requires infrastructure	Uses available data
Higher cost/dependency	Lower cost
Harder to deploy	More feasible MVP

This is **excellent PM evidence**.

---

10. MVP Definition

Now introduce the actual product.

I would show the product as **three layers**:

Commuter experience

- Route discovery
- Timetable
- ETA
- Saved routes
- Reminders
- Multilingual experience

Operations

- Schedule/timetable management
- Operational updates
- Role-based workflows

Intelligence

- ETA calculation
- Scheduling/optimization
- Route/timetable data processing

Then show screenshots from your actual product.

This section should be visual-heavy.

---

11. Prioritization

This is currently one of the biggest gaps we need to reconstruct.

We need to explain:

> **Why these features? Why not others?**

Even if you didn't use a formal RICE/MoSCoW framework in 2018, we can reconstruct your actual reasoning.

For each major feature:

User problem → expected value → feasibility → dependency → decision

Don't retrospectively pretend you used a framework if you didn't.

That's important.

---

12. Product / UX Journey

Show one commuter journey:

I need to travel
↓
Where is my destination?
↓
Which route should I take?
↓
When does the bus arrive?
↓
How long will I wait?
↓
Can I save/remind myself?
↓
Complete journey

Then map the product features against that journey.

This demonstrates **customer-centric thinking** better than a list of features.

---

13. Technical & Cross-functional Execution

Keep this relatively short.

Your technical background is useful here, but **don't let the engineering overwhelm the PM story.**

Mention:

- Android application.
- Data model for 90 routes.
- Data cleaning/parsing.
- JSON model.
- Java processing.
- Firebase/Node.js/cloud functions where relevant.
- Scheduling/ETA logic.

Your resume materials support that you translated requirements into implementation and have a strong engineering foundation.

But the framing should be:

> **How I worked with engineering**

rather than:

> **Look how much code I wrote.**

---

14. Leadership & Stakeholder Management

This deserves its own section.

Explain your role in the 7-person team.

Your contribution

- originated the opportunity,
- created alignment around the idea,
- allocated responsibilities,
- coordinated UI/technology/research work,
- wrote requirements,
- interacted with external stakeholders,
- coordinated execution,
- helped secure college funding,
- drove toward deployment.

Team contribution

Clearly separate what your teammates did.

This is extremely important because **we must never make the portfolio imply that you personally built everything.**

---

15. Validation & MVP Feedback

Show what happened when you presented the solution to stakeholders.

We should document:

- What stakeholders liked.
- What concerns they raised.
- What changed because of feedback.
- Whether they considered it promising.
- What was required for launch.

This is currently a section where I need to extract more detail from you before finalizing.

---

16. Metrics & Expected Impact

This needs **very careful treatment**.

You have several numbers floating around:

- 70,000+ commuters
- 15% revenue improvement target
- 20% waiting-time reduction target
- 90 routes
- 2-minute data-processing improvement

We need to categorize them.

Example:

Problem scale

> 70,000+ daily commuters.

Product target

> Proposed 20% reduction in waiting time.

Business-case target

> Proposed 15% revenue improvement.

Technical result

Reduced data-processing time by ~2 minutes.

Actual product outcome

Deployable MVP completed; public launch did not occur.

That distinction is **essential**.

Do not write:

"Reduced waiting time by 20%"

unless you actually measured it.

Don't write:

"Served 70,000 users"

because the product wasn't publicly launched.

This credibility will matter enormously to senior PM reviewers.

---

17. The Launch That Didn't Happen

This should be one of the strongest sections.

We built the MVP. But the market moved underneath us.

Explain:

- You were preparing for a beta/Play Store launch.
- MSRTC requested additional documentation/budgeting.
- Further approval from Mumbai headquarters became a requirement.
- Meanwhile, the organization was moving toward transferring city bus operations.
- Eventually the project remained an academic project rather than a publicly launched product.

Don't frame this as:

"MSRTC killed our product."

Frame it as:

> **Our product encountered organizational misalignment at the point of adoption.**

That's much more mature.

---

18. What Went Wrong?

This is where your PM maturity can shine.

Break the failure into:

Product feasibility

Could we technically build it?

Yes.

User value

Did commuters have a meaningful problem?

Yes.

Business value

Could it potentially improve operations/revenue?

The proposal targeted this; actual impact was not measured.

Organizational readiness

Was the intended customer ready and incentivized to adopt it?

Ultimately, no.

That is a fantastic framework for explaining what happened.

---

19. What I Would Do Differently Today

Your answer is already strong.

Today you would:

1. Map the ecosystem before building.
2. Identify who actually owns the problem.
3. Understand incentives of MSRTC and NMC.
4. Validate adoption feasibility.
5. Secure executive sponsorship.
6. Identify critical data/infrastructure dependencies.
7. Define a pilot with explicit success metrics.
8. Establish an adoption/ownership path before development.
9. Only then invest heavily in building.

This directly connects your 2018 experience to your current PM maturity.

---

20. Product Lessons

I'd keep this to **3–5 memorable principles**, not 15 generic lessons.

1. Solve the right problem before solving it well.

2. Constraints are product inputs.

The GPS → heuristic ETA pivot proves this.

3. A product doesn't exist in isolation.

Users, technology, business incentives, regulation and organizational realities all matter.

4. Validate adoption, not just desirability and feasibility.

This is perhaps your most important retrospective lesson.

5. Great PMs reduce uncertainty before committing resources.

This connects beautifully with how you actually worked.

---

21. Closing: What NBA Taught Me About Product Management

End the case study by bringing it back to **you**.

Something along these lines:

> NBA started as an attempt to solve a problem I experienced every day. It became my first experience of taking a product from an ambiguous problem to a deployable MVP—and my first lesson that building something useful is only one part of product management. A product also needs the right incentives, stakeholders, infrastructure and environment to survive.
>
> That experience is a major reason I think about products as systems rather than collections of features.

That's the ending I want a recruiter to remember.

Sounds good, let us start the next section. I'll tell you the section name and you can give me the content (not more than 1 page). All the content should be concise and simple to read. You can ask me any clarifying Qs, give me quality suggestions to improve the richness of the content.

Nashik Bus Assistant

Making Nashik's public bus system more predictable for daily commuters

In 2018, I led a 7-member team to design and build a low-cost digital public transport system for Nashik, addressing unreliable bus schedules, long commuter wait times, and limited access to bus information. We explored the solution with MSRTC and pitched it as a potential public transport technology solution under Nashik's Smart City initiative.

My role	Product Lead / Scrum Master
Team	7 members
Timeline	2018–2019
Domain	Public Mobility / Civic Tech
Problem scale	70,000+ daily commuters
Stakeholders	MSRTC, NMC, depot managers, drivers, conductors & commuters
Product	Commuter Android app + Conductor/Controller Android app + scheduling system
Outcome	Deployable MVP; public launch was not completed due to organizational and adoption constraints

[Watch MVP Demo →]

See the commuter application in action.

[Product snapshots]

Conductor/Controller Android app

My role

I led the product from **problem discovery to MVP**, working across user research, stakeholder engagement, product requirements, solution design, team coordination, and key product decisions.

The Story: Why I Built This

A newspaper headline turned a personal frustration into a product opportunity

In 2018, while taking an NPTEL course on **Object-Oriented Analysis and Design**, I was learning to approach software as systems built around users and their needs. When I asked my professor what makes a good final-year project, he suggested looking beyond academic papers and exploring real-world problems.

Around the same time, I read about the financial struggles of Nashik's city bus service. The problem immediately resonated with me—I had spent nearly three years dealing with long, unpredictable waits for buses, often without knowing when the next bus would arrive.

I had about 1.5 months before the next academic year began, so I decided to investigate whether technology could make the experience more predictable for commuters. I brought the idea to my friends, and a team of seven began exploring the problem.

What started as a personal frustration became an opportunity to understand and improve a city-wide public transport system.

A commuter problem inside a financially constrained and changing transport ecosystem

When we began working on Nashik Bus Assistant in 2018, the city bus system was facing a combination of **financial pressure, declining fleet capacity, poor commuter experience, and an ongoing transition in operational responsibility**.

This meant the problem was bigger than simply making bus timings available digitally.

The ecosystem we were entering

🚌 Commuters — *The poor experience problem*

Nashik's daily commuters faced:

- Long and unpredictable waiting times
- Low bus frequency
- Irregular schedules
- Difficulty knowing when the next bus would arrive
- Limited information about routes and nearby bus stops
- No convenient digital channel for accessing bus information or raising concerns

As a daily commuter, I had experienced this problem firsthand for several years.

⚙️ MSRTC — *The operational & financial problem*

MSRTC was operating Nashik's city bus service under significant financial pressure.

According to the contemporaneous reporting we reviewed:

- City bus losses were reported at **₹17.72 crore in 2016–17** and **₹11.8 crore in 2017–18**.
- The city fleet had reduced from **265 buses in January 2017 to 110 buses by April 2018**.
- MSRTC was seeking ways to reduce the financial burden associated with city bus operations.

The reduction in fleet capacity directly affected commuter experience while creating a strong incentive for MSRTC to improve operational efficiency and reduce losses.

Source: Times of India — MSRTC to continue with 110 buses for city

---

🏙️ NMC — *The ownership & sustainability problem*

At the same time, there was an ongoing debate between **MSRTC and Nashik Municipal Corporation (NMC)** over responsibility for city transportation.

NMC was being pushed toward taking greater responsibility for the city's public transport system, while working through questions around:

- Financial sustainability
- Operational capability
- Ownership
- Future operating model

This created an important organizational constraint: **the organization operating the service was not necessarily going to be the organization responsible for its future.**

The strategic tension

This created three different sets of incentives:

Stakeholder	Primary need
Commuters	Better availability, predictability and journey information
MSRTC	Reduce operational burden, improve efficiency and address financial losses
NMC	Build a sustainable, technology-enabled city transport system

And this is where we saw a potential role for technology.

Our opportunity hypothesis

> **Could we use a low-cost digital system to improve the commuter experience while helping the transport authority operate its bus network more efficiently?**

Our pitch therefore went beyond a commuter-facing application.

The project proposed:

For commuters
→ Better access to routes, timings, fares and bus information

For transport operations
→ Better timetable management and bus scheduling

For the governing body
→ Better communication with commuters and potential improvements in operational and financial performance

Problem Discovery

We didn't start by building an app. We started by understanding the system.

Our initial hypothesis was:

> **Commuters struggle because bus schedules are unreliable and there is no digital system to provide timely bus information.**

Before deciding what to build, our 7-member team spent time experiencing and investigating the problem from different perspectives.

1. We observed commuters in their real environment

All seven of us were regular city-bus users. We deliberately traveled on **different routes and buses** and spoke with commuters during their journeys, at college boarding points, bus stops and different city-bus depots.

Across these interactions, we conducted **50+ conversations with daily commuters** at different points in time.

We explored questions such as:

- How long do you typically wait for a bus?
- How do you decide which bus to take?
- How do you know when your bus will arrive?
- What happens when a bus is late or full?
- What information would make your journey easier?

This helped us validate that our own frustration was not an isolated experience.

2. We investigated the operational side

We also spoke with **depot managers, controllers, conductors and other transport stakeholders** to understand how the bus system actually worked.

We looked at:

- How routes and schedules were managed
- How bus timings were maintained
- How operational information flowed
- How buses were managed across depots
- What data was available to us

This was important because a commuter-facing product would depend on the underlying operational system.

3. We collected and studied operational data

We obtained route and timetable information from **MSRTC** and examined how the available data could be structured for a digital product.

We eventually cleaned and organized information covering **90 bus routes** for use in the system.

4. We looked beyond Nashik

We studied public transport systems in cities such as **Indore, Mysore and Bengaluru** to understand how other cities approached route information, scheduling and commuter experience.

What changed in our understanding?

Our initial assumption was:

> **"Build a real-time bus tracking system."**

But discovery exposed several assumptions we hadn't validated:

We initially assumed	We discovered
GPS could power real-time tracking	GPS infrastructure wasn't available to us
Required bus-stop data would be available	Complete coordinates were not available
The problem was primarily commuter-facing	The solution also depended on transport operations
We could design around the ideal solution	We needed to design around actual infrastructure and data constraints

The key insight

The problem wasn't simply "commuters need GPS tracking."

The underlying need was **predictability**—helping commuters make better decisions about their journey using reliable information.

At the same time, the solution had to work within the **data, infrastructure and operational constraints of MSRTC**.

This changed our next question from:

> *"How do we build a real-time bus tracking app?"*

to:

> **"What useful experience can we create with the data and infrastructure that actually exist?"**

That question led to our first major product pivot.

5. What We Learned

Our research changed our understanding of the problem and directly influenced how we designed the product.

1. The core commuter need was predictability, not simply more buses

What we observed: Commuters struggled with irregular schedules and uncertainty about when a bus would arrive.

What it meant: While we could not immediately change fleet capacity or frequency, better information could help commuters make more informed travel decisions.

Product implication: We focused the commuter experience on **route, timing and bus availability information** rather than trying to solve the entire public-transit capacity problem.

---

2. Data availability was a product constraint

What we learned: Route and timetable information was controlled by the transport organization, and the data was not available in a format ready for a digital product.

What it meant: Data collection, cleaning and maintenance were part of the product problem—not just backend engineering.

Product implication: We structured information covering **90 bus routes** and designed the system around the data that could realistically be maintained.

---

3. The ideal GPS solution was not feasible in our environment

Our initial concept was a GPS-enabled, real-time tracking experience.

We discovered that we did not have **GPS infrastructure or complete bus-stop coordinates** required for that approach.

Product implication: Instead of abandoning the problem, we pivoted toward a **lower-cost heuristic approach to estimate bus arrival information using available data and infrastructure.**

This was one of our earliest examples of balancing **user value with technical and operational feasibility.**

---

4. The product was a two-sided system

We initially thought primarily about the commuter experience.

But we realized that reliable commuter information depended on the people maintaining the underlying operational data.

This led us to design two connected experiences:

- **Commuter App:** Help users find routes, buses, timings and travel information.
- **Controller/Conductor App:** Enable transport staff to manage operational information that feeds the commuter experience.

Product implication: We had to design the system around both **user needs and operational workflows.**

---

5. Product value does not guarantee adoption

Our biggest learning came later.

We had built an MVP that addressed a genuine commuter problem and received initial interest from stakeholders. However, when we approached MSRTC for deployment, the organization's broader strategic situation became critical.

MSRTC was under significant financial pressure and was moving toward transferring responsibility for Nashik's city bus operations.

What I learned: A product can be technically feasible and solve a real user problem, yet still fail to get adopted if it does not align with the **organization's incentives, priorities and strategic direction.**

NBA wasn't a single-user product. The commuter experience depended on information and workflows controlled by transport staff, while the broader product needed support from the organizations responsible for city transportation.

We therefore thought about the product across **three stakeholder groups**.

🚌 1. Commuters — Primary end users

Who: Daily city-bus commuters, including students and other regular passengers.

Needs:

- Find the right route
- Understand bus timings
- Reduce uncertainty while waiting
- Know which buses serve their journey
- Access information without depending on informal sources

Our product response:
The **Commuter App** was designed around route discovery, bus timings and accessible journey information.

⚙️ 2. Controllers & Conductors — Operational users

Who: MSRTC staff involved in managing and operating city bus services.

Needs:

- Maintain operational information
- Manage user access
- Maintain bus/timing information
- Provide reliable information that could reach commuters

Our product response:
We designed a separate **Controller/Conductor App** to support operational workflows and maintain the information powering the commuter experience.

This was an important realization: **a good commuter experience depended on making the operational side of the system work as well.**

🏙️ 3. MSRTC & NMC — Institutional stakeholders

MSRTC
Responsible for city bus operations during our project. Their priorities included operational efficiency and reducing the financial burden of city operations.

NMC
The civic body involved in the transition of responsibility for city transportation and the broader Smart City initiative.

Their needs:

- A feasible and affordable technology solution
- Better visibility and communication
- More efficient transport operations
- Potential improvement in financial performance

We therefore positioned NBA as more than a commuter application—it was a **potential technology layer connecting commuters with the transport operation.**

The key product insight

The **commuter was the primary beneficiary**, but the **transport organization was a critical enabler**.

That meant we couldn't optimize the product only for the end-user experience. We also had to consider the workflows, data and incentives of the organization that would operate and potentially adopt it.

Designing for the user meant understanding the ecosystem around the user.

Product Vision

Make city bus travel more predictable through accessible, low-cost technology

Our vision for **Nashik Bus Assistant (NBA)** was to make public bus travel more predictable for Nashik's daily commuters by providing accessible information about **routes, bus timings and journeys**.

At the same time, we wanted to create a digital system that could support the transport organization's operational workflows and improve how information flowed between **transport staff and commuters**.

The product vision

> **Create a low-cost digital layer that connects public transport operations with commuters, making bus information easier to access and journeys more predictable.**

This led us to design two connected product experiences:

🚌 Commuter App

- Search for routes and destinations
- View relevant bus timings
- Access journey information more easily

⚙️ Controller/Conductor App

- Manage operational information
- Maintain bus and timing data
- Support the information flow powering the commuter experience

What we were ultimately trying to achieve

For commuters
→ Reduce uncertainty and make journeys easier to plan

For transport operations
→ Improve how route and timing information was maintained and communicated

For the broader transport system
→ Introduce a practical, low-cost digital layer without requiring a complete overhaul of existing infrastructure

> **Our vision was not to build another bus-tracking app. It was to make an existing public transport system more predictable and accessible through technology.**

We changed the solution—not the problem we were trying to solve

Our initial solution concept was a **GPS-enabled real-time bus tracking system**.

It seemed like the natural way to address unpredictable bus arrivals. But before committing the team to that approach, we tested its key dependencies.

We found that:

- GPS infrastructure was not available across the network.
- Complete bus-stop coordinates were unavailable.
- The team needed to evaluate feasible backend/cloud technologies for the MVP.

Rather than continuing with a solution whose core dependencies we couldn't satisfy, **I asked the team to investigate the constraints before we committed to the architecture.**

From GPS tracking to a low-cost ETA approach

The key product decision was:

> **Don't change the user problem because the first solution isn't feasible. Change the solution.**

We explored a **heuristic ETA approach** that could use available route and timetable information to estimate bus arrival information without depending on GPS infrastructure.

	Initial concept	MVP direction
Core mechanism	GPS-based tracking	Heuristic ETA
Data dependency	GPS + complete stop coordinates	Available route/timetable data
Infrastructure	New GPS infrastructure required	Lower infrastructure dependency
Goal	Improve journey predictability	Same

Designing the complete system

We also recognized that the commuter experience depended on operational information being maintained behind the scenes.

So the MVP consisted of two connected experiences:

🚌 **Commuter App**
Search routes, find buses and view timing information.

⚙️ **Controller/Conductor App**
Manage users and maintain operational/timing information.

A backend layer connected the two sides and processed the route and timetable data.

Prioritization

We prioritized the foundation before the optimization layer

Our original product vision was broader than the first MVP. We wanted NBA to improve both the **commuter experience** and **bus operations**, including eventually optimizing bus schedules using a genetic algorithm.

Rather than trying to deliver everything simultaneously, we structured the product into stages.

What we prioritized first

1. Commuter information — Must have

The first priority was giving commuters access to the basic information they lacked:

- Available buses
- Routes
- Timings
- Distance and travel information
- Fare information
- Nearby bus stops

This formed the core value proposition of the Commuter App.

2. Operational data management — Must have

The commuter experience depended on accurate operational information.

We therefore prioritized the **Controller/Conductor App** and the underlying data layer to allow transport staff to provide and maintain timely information.

3. Bus scheduling optimization — Later

The longer-term vision included a genetic-algorithm-based scheduling system that would use user travel details such as source, destination, travel timings and travel frequency to generate optimized bus schedules.

However, this was positioned as the **next milestone after the initial application module**, rather than something we needed to complete before proving the core product experience. The project plan explicitly shows the Android module as completed and the bus-scheduling algorithm as the next milestone.

Our prioritization logic

We effectively followed this sequence:

Establish the information foundation
↓
Enable the commuter experience
↓
Enable operational updates
↓
Validate the core product
↓
Add advanced scheduling optimization

The principle was simple:

> **First make the existing transport information more accessible and usable; then build intelligence on top of that foundation.**